

Suggested problems: 1-6, 8-12, 14, 15

Problem 1: Find the cumulative distribution function of a random variable $X \sim \mathcal{Ber}(p)$.

Problem 2: Find the cumulative distribution function of a random variable $X \sim \mathcal{Bin}(3, 1/5)$.

Problem 3: Let X be a continuous random variable with density function

$$f(x) = \begin{cases} 2x & : 0 < x < 1, \\ 0 & : \text{else.} \end{cases}$$

Find its cumulative distribution function.

Problem 4: The length of time to failure (in hundreds of hours) for a transistor is a random variable X with cumulative distribution function given by

$$F(x) = \begin{cases} 1 - e^{-x^2} & : x \geq 0, \\ 0 & : \text{else.} \end{cases}$$

Compute the probability $P(X > 1 \mid X \leq 2)$.

Problem 5: Suppose $X \sim \text{Bin}(3, 1/5)$. Compute the following quantities.

(a) The 95-th percentile.

(b) The 0.8-th quantile.

(c) The median.

(d) The 0.1-th quantile.

Problem 6: For the random variable X in Problem 4, compute a formula for the quantile function.